WHAT CHARACTERISTICS OF STUDENT AFFECT THEIR FINAL GRADE IN MATHEMATIC?

This is a practical summary of an analysis of Academic performance in Mathematics course in the final year of high school in two Protuguese schools.

LET’S HAVE A LOOK AT THE SAMPLE!

- 395 students (52.66% female students)
- 67% passed Mathematics (final grade 10 or higher)
- 88% attended Gabriel Pereira school

Important note:

The sample is heavily weighted toward one school, so the overall pattern may reflect that school more strongly.

THE CLEAREST SIGNAL: PAST CLASS FAILURES

- **≈ 2 points** lower final grade associated with each additional previous failure

What this could mean for schools

- Previous failures may help identify students who need additional academic support.
- A useful response is early, supportive follow-up—not labelling students by their past results.
- The analysis identifies a pattern; it does not establish why it occurs.

OTHER FINDINGS NEED A CAREFUL READING

What the models found

- Male students had higher estimated grades and pass odds in this sample, after adjustment.
- Higher-education plans were linked to higher grades, but not clearly to passing.
- Study time and absences were not statistically significant at the 5% level.

What not to conclude

- That sex itself causes differences in achievement.
- That study time or absences do not matter in general.
- That a non-significant result proves there is no relationship.

WE SHOULD KEEP IN MIND

- Generalizability of the findings is limited, since they are based on only two Portuguese schools.
- Privacy and potential bias can influence our findings.
- Correlation does not always mean causation.
- A predictive models should not be used as the sole basis for decisions affecting students' educational opportunities.

USE THE FINDINGS AS A PROMPT—NOT A VERDICT

Check in early

Use previous failures as one signal for a supportive conversation or additional help.

Look beyond grades

Consider learning needs, motivation, family circumstances, teaching quality, and access to support.

Validate locally

Check whether similar patterns appear in other schools and in newer cohorts.